Loyal Spark

Loyalty-as-a-Service on Base - for merchants, customers, and AI agents

Sentient Foundation Open Source AGI Grant Application - Supporting Document

Executive summary

Loyal Spark is Loyalty-as-a-Service (LaaS) on Base Mainnet. Merchants launch full onchain loyalty programs (ERC-20 token, minting, rewards, vouchers, CRM) through a web portal - without building Web3 infrastructure from scratch. Customers hold points in their own wallets. Developers and autonomous AI agents use the same backend via REST API and MCP (Model Context Protocol).

The codebase is MIT-licensed on GitHub. Smart contracts are live on Base. We expose open agent discovery (agent.json, OpenAPI, llms.txt, 13 Markdown skills) and support wallet-only agent registration (SIWE) with no dashboard required. Paid agent corridors use x402 micropayments (USDC on Base).

At a glance

- Live on Base Mainnet (Chain ID 8453)
- Merchant portal + customer portal (Privy, SIWE, embedded wallets)
- Agent API: 23 REST routes + 32 merchant MCP tools + 14 recipient MCP tools
- x402 and MPP pay-per-call gateways
- Capacitor native apps scaffolded (iOS / Android - grant target: store release)
- Open source: github.com/aspekt19/unboxed-loyalty-spark (MIT)

What the $25,000 grant unlocks (4-6 months)

Bootstrapping, I can keep mainnet running - but I cannot properly ship mobile apps, gasless UX, and open ecosystem work at the same time. This grant funds focused delivery:

1. App Store + Google Play

Ship two Capacitor apps already scaffolded in the repo: Loyal Spark Business (merchants) and Loyal Spark (customers). Includes store assets, QA, review cycles, and dev account costs. Goal: loyalty that normal businesses can adopt as a mobile app, not only a website.

2. Paymaster on Base (gasless transactions)

Integrate a Paymaster so merchants and customers can mint, redeem, and transfer without holding ETH for gas. Wire through existing Privy + embedded wallet flows. Document the integration pattern in the MIT repo for other builders.

3. Open agent layer

- Expand the public skills bundle (merchant + holder end-to-end flows)
- Reference MCP client configs (Cursor, Claude Desktop)
- Polish SIWE agent registration (lsk_ API keys without UI)
- x402 paid MCP/API examples; target 10+ unique external paying agent wallets

4. Merchant pilots + live infra

- Onboard 2-3 pilot merchants: deploy, mint, voucher - web + mobile + gasless
- 6 months production infra (RPC, Supabase, monitoring) so API/MCP stay up for integrators

Without the grant: mainnet stays up, but mobile, Paymaster, open docs, and merchant pilots stall - hurting anyone building on our open repo.

Problem

- Legacy loyalty is closed: customers do not truly own points
- Small businesses cannot afford custom Web3 builds or $99-499/mo legacy SaaS with revenue cuts
- Going onchain often ends as a bare token with no vouchers, tiers, or operations
- AI agents need the same economic layer - issue value, reward users, settle payments - but lack a ready stack

Solution - Loyalty-as-a-Service

One platform on Base where any merchant can launch a full loyalty program in minutes - same infrastructure for everyone, not a custom build per brand.

Merchants (portal)	Deploy token, mint, rewards, vouchers, team, CRM
Customers (wallet)	Own tokens, redeem, transfer, P2P escrow marketplace
Agents (API + MCP)	Automate programs; pay per call via x402 or API keys (lsk_ / rwk_)

Why now

- Cheap L2 transactions and USDC on Base
- Wallet abstraction (Privy) lowers the bar for non-crypto merchants and customers
- MCP and x402 make agent-to-agent and pay-per-request commerce practical
- Missing piece: an open, composable loyalty layer - not another one-off token per project

Product - live today

Merchant flow

- Deploy ERC-20 loyalty token via factory contract
- Mint to email, phone, or wallet; vouchers, tiers, RFM, automation
- Invite team (branches, employees) and AI agents (scoped API keys)

Customer flow

- Receive tokens to embedded or external wallet
- Redeem rewards; trade on P2P escrow marketplace
- Holder-side API (rwk_) and recipient MCP for agent-driven redemption

Agent / developer surface

- REST agent-api + loyalty-mcp (merchant) + recipient-loyalty-mcp (holder)
- Autonomous registration: SIWE message, no web login required
- CDP MPC server wallets for autonomous onchain operations
- x402-gateway and mpp-gateway for pay-per-request access

What's open (MIT on GitHub)

- Full application and Edge Function source
- 13 step-by-step agent skills (skills/loyal-spark/)
- MCP examples, OpenAPI, agent.json, llms.txt
- Contract ABIs, addresses, autonomous agent registration docs

If we shut down tomorrow: merchants lose a working LaaS stack; customers lose live programs; developers lose running API/MCP and x402 endpoints. The repo can be forked, but production continuity would break.

Smart contracts - Base Mainnet

LoyaltyTokenFactory	0x5F3DdBa12580CFdc6016258774cCc19C4250dA80
LoyalSparkERC20 (impl)	0xe6BA426C9c51281B929a17444De02c65815E27C3
LoyaltyTokenEscrow (P2P)	0xA569C95AfC1BCF381c48BcF336ED9D2c014bcdDF
Builder Code (ERC-8021)	bc_wdmnog7m

Team

Solo founder - built Loyal Spark as LaaS on Base, then added the agent layer (REST, MCP, x402, CDP wallets) on the same backend. Shipped contracts and product on mainnet. Ships in public on GitHub (MIT).

Grant request

- Track: Grant (non-dilutive)
- Amount: $25,000 USD
- Focus: mobile store release, Paymaster, open agent docs, merchant pilots

Demo & trial links

Website:	https://loyalspark.online
Merchant portal:	https://loyalspark.online/merchant
For agents / onboarding:	https://loyalspark.online/for-agents
API documentation:	https://loyalspark.online/api-docs
Agent manifest:	https://loyalspark.online/.well-known/agent.json
GitHub (MIT):	https://github.com/aspekt19/unboxed-loyalty-spark

Contact

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